

Assignment 2:

Member-1 Name: _____ Roll Number: _____

Member-2 Name: _____ Roll Number: _____

Section: _____

Task 1: Data Uploading & Cleaning (Total: 20 marks)

Criteria	Marks
Data correctly loaded, primary key uniqueness ensured, and referential integrity maintained.	10
Data cleaning steps and modifications documented concisely.	10
Subtotal:	20

Task 2: Data Retrieval (SQL Analysis) (Total: 115 marks)

A. Order Analysis (Total: 30 marks)

Query Requirement	Marks
1. Percentage of orders delayed beyond the estimated date	3
2. Peak months of order delays	3
3. Which state experiences the highest order delays?	4
4. How many orders are still in “pending” status for each year?	3
5. What is the average delay duration per seller?	4
6. Impact of shipping costs on order delays (avg shipping cost for delayed vs. on-time orders)	4
7. Which product category experiences the most order delays?	4
8. How do the number of items per order affect delays? (avg items for delayed vs. on-time orders)	5
Subtotal:	30 Marks

B. Customer Analysis (Total: 30 marks)

Query Requirement	Marks
1. Percentage of customers with only one order	3
2. Top five cities with the most repeat customers	4
3. Average order price per state	4
4. Top ten customers with the highest number of orders placed	3
5. Which customers have the longest average delivery time?	4
6. How does customer order frequency change over time? (avg orders per customer per year)	4
7. Top 5 customers who spent the most in 2017	5
8. Which customers have the highest order cancellations?	3
Subtotal:	30 Marks

C. Product Analysis (Total: 31 marks)

Query Requirement	Marks
1. Most profitable product category per state	5
2. Peak hours for order placements per product category	4
3. Top 5 product categories with the highest number of delayed orders	4
4. Impact of product price on sales volume (avg price vs. total sales)	4
5. Products most frequently bought together	5
6. Total revenue per product category	3
7. Average review score per product category	3
8. Top 5 products based on total sales revenue	3
Subtotal:	31 Marks

D. Seller & Shipment Analysis (Total: 24 marks)

Query Requirement	Marks
1. Return rate per seller	4
2. Sellers selling the most expensive products on average	3
3. Profit margin per seller	4
4. Find the average shipping cost (freight_value) for delayed vs. non-delayed orders.	3
5. Number of delayed shipments in 2017	3
6. Correlation between shipping cost and delivery speed (avg freight and delivery days for delayed vs. on-time shipments)	4
7. Total freight cost per seller	3
Subtotal:	24 Marks

Task 3: Presentation (Total: 10 marks)

Requirement	Marks
Export all query results (with >10 rows) to CSV files and generate clear graphs (in Excel/Google Sheets).	5
Provide a well-organized PDF report with proper title page, sections, and screenshots of graphs (with brief insights, no SQL code).	5
Subtotal:	10 Marks

Total Marks: 145

Obtained Marks: